Cerebellum: Cram Sheet (NSC 4351)

Slides + lecture recording. **Red = said in lecture: stressed, "will test," or not on the slides.** Rule of 3s everywhere: 3 lobes, 3 zones, 3 nuclei, 3 peduncles, 3 layers, 3 syndromes.

The whole lecture in 6 lines

- **Cerebellum corrects movement; it never starts it.** Lesion = clumsy, wobbly, badly timed. Strength and sensation normal.
- **Signs are IPSILATERAL** (double cross). Everything else in the brain is contralateral.
- **Lobes:** anterior = legs, posterior = arms, flocculonodular = balance + eyes.
- **Zones:** vermis = trunk, paravermis = limbs, lateral = hands + planning.
- **Circuit:** all inputs excite; only Purkinje output inhibits. **Cerebellum = the brake.**
- **Tremor:** intention = cerebellum. Rest = basal ganglia.

🔴 What he said he will test

- **Picture: find the DENTATE.** Big crinkled ribbon, far lateral, "looks like teeth." He will NOT ask emboliform vs globose.
- **"Learn this picture and you're home free":** the homunculus on the spinocerebellum. Trunk on vermis (midline), limbs on paravermis, hands huge and lateral.
- **All inputs excitatory, only Purkinje output inhibitory.** How does anything get out? Purkinje inhibits the deep nuclei, and the nuclei are the output. The job is to step on the brake.
- **Climbing vs mossy:** direct and powerful vs indirect and weak. Far fewer climbing fibers because they are too powerful.
- **Gait is the gold standard.** Weakness, sensory loss, cerebellar disease, intoxication all show in the walk.
- **Why intention tremor is worst near the target** (see Syndromes).
- **Rest vs intention tremor.** No tremor at rest in cerebellar disease because nothing is moving, so nothing to coordinate.
- **Cerebellar signs from damaged INPUT:** B12 deficiency kills the spinocerebellar tracts; patient looks cerebellar but the cerebellum is fine. Same for a pontine / MCP lesion.
- **Alcohol = 3 stories:** intoxication, withdrawal, chronic Purkinje loss.
- **Posterior lobe syndrome = arms and speech.** Flocculonodular = trunk. Anterior = legs.
- Bonus brainstem quiz coming; points added to Exam 2.

1. Anatomy and the rule of 3s

Landmarks: vermis (midline worm), hemispheres, primary fissure (anterior/posterior split), nodulus (midline, under vermis near 4th ventricle), flocculus (lateral tuft), tonsils (lowest part, herniate in Chiari), folia, arbor vitae (white matter tree on sagittal cut). Flat "bow tie" diagrams are the folded cerebellum spread out by an artist; translate for the picture quiz.

Transverse cut = 3 lobes

Lobe	Job	Damage
Anterior (front of primary fissure)	**Legs**	Lower-limb ataxia
Posterior (biggest)	**Arms**	Intention tremor, dysmetria, dysdiadochokinesia, scanning speech
Flocculonodular (oldest)	**Eye movements, posture**	Truncal ataxia, titubation, wide base

Hook: front = feet, back = arms, bottom = balance. Posterior lobe is biggest because hands need the most coordination. Flocculonodular is the oldest part; it is where the common childhood midline tumor (medulloblastoma) arises (beyond this class).

Longitudinal cut = 3 zones

Zone	Job	Nucleus	Input
Vermis (midline)	**Trunk posture**	Fastigial	Spinal cord proprioception
Paravermis (intermediate)	**Gross ipsilateral limb movement** (walking)	Interposed (emboliform + globose)	Spinal cord, trigeminal, red nucleus, visual, auditory
Lateral (cerebrocerebellum)	**Skilled ipsilateral hand movement** + planning + cognitive sequences	Dentate	Motor and association cortex via pons

Other names: vestibulocerebellum = flocculonodular; spinocerebellum = vermis + paravermis; cerebrocerebellum = lateral. Vermis lesion: cannot sit straight. Paravermis: trouble coordinating arms and legs, walking. Lateral: skilled hand tasks. Diseases pick regions: genetic one place, alcohol another, childhood tumors midline.

2. Wiring: in, compute, out

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Inputs → cerebellar CORTEX → DEEP NUCLEI → output
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The exact reverse of the cerebrum: cerebrum relays through deep structures to end in cortex; cerebellum puts input into cortex first and sends output from the deep nuclei.

Deep nuclei, lateral to medial: Don't Eat Greasy Food

Nucleus	Pairs with	Job
Dentate	Lateral zone	Fine limb movement; biggest; out via SCP. The one on the picture quiz.
Emboliform + Globose = interposed	Paravermis	Ipsilateral limbs, grosser movement. Not asked apart.
Fastigial	Vermis / flocculonodular	Posture. Vignette: truncal ataxia + titubation, limbs fine = fastigial.

3 peduncles (reason it out "like the engineer")

Peduncle	Dir.	Carries
Inferior (ICP)	IN	Everything from BELOW: inferior olive (climbing fibers), spinal cord (spinocerebellar tracts, Clarke's nucleus), vestibular nuclei
Middle (MCP)	IN, **main**	Cortex → pontine nuclei → cross → cerebellar cortex. Huge because nearly all of cortex must talk to the cerebellum before any coordinated move (red bundle on tractography vs tiny yellow spinal bundle).
Superior (SCP)	OUT	Deep nuclei → cross → **VL thalamus** → motor/premotor cortex; also → red nucleus → inferior olive → back as climbing fibers

Hook: Superior Sends, Middle is Main input, Inferior takes In the rest. **Lesion of the pathway gives cerebellar signs without cerebellar damage:** pons, cerebral peduncle, MCP, or B12 deficiency (subacute combined degeneration wipes dorsal and ventral spinocerebellar tracts). Red nucleus damage = rubral tremor (coarse wobble lifting a water bottle).

Why signs are ipsilateral: the double cross

Right cerebellum → SCP crosses → left thalamus, left motor cortex → corticospinal crosses in medulla → RIGHT body. Two crosses cancel. **Right cerebellar lesion = right limb ataxia.** (Right cortex lesion = left weakness.)

3. Microcircuit (3 layers, vs 6 in cerebrum)

Layer	Contents
Molecular (outer)	Parallel fibers, Purkinje dendrites, stellate + basket cells
Purkinje	Single row of huge Purkinje cell bodies
Granule (inner)	Granule cells (most numerous neuron in the brain), Golgi cells

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Climbing fiber (inferior olive ONLY) → excites Purkinje
Mossy fiber (all other afferents) → excites granule cell
Granule cell → parallel fiber → excites Purkinje
Purkinje → INHIBITS deep nuclei (GABA)
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Bank analogy: climbing fiber walks straight into the vault: ONE fiber wraps the ENTIRE dendritic tree of ONE Purkinje cell and controls its whole firing pattern. Mossy fibers go through middlemen (granule cell → parallel fiber → one weak synapse on each of many Purkinje cells); weak modifiers but enormous numbers. Fewer climbing fibers because they are too powerful. **Driving analogy:** cortex says "go," cerebellum says "slow down, slow down, now turn." Without it you overshoot, swerve back, overshoot again = dysmetria. Picture: red/green = excitatory (mossy, climbing, parallel), blue/black = inhibitory (Purkinje, basket, stellate, Golgi).

4. Functions and the correction loop

3 jobs: control posture; correct rapid movements started by cortex; motor learning.

1. Cortex sends the **intention** (via pons, MCP)
2. Proprioceptors + vision report **body position** (via ICP)
3. Cerebellar cortex computes the **best way**
4. **Blueprint** back to motor cortex (SCP, VL thalamus)
5. Movement happens; errors → new plan to cerebellum for **correction**

"Cortex: we're making this move, you monitor." Cerebellum predicts the endpoint ("at this speed you'll land over here"), yells "you're off, give me a new plan," cortex corrects, repeat, fast. **Stored programs:** things done thousands of times (words, counting in your native language, practiced skills) live in the cerebellum as finished programs; a second language counts twice as slow because the program isn't written yet. Spinocerebellar ataxia patient: can do each piece, cannot string the sequence. **You need proprioception to move on purpose:** "biceps" touches the nose only if the arm starts down; if it starts up behind you, "triceps" first. Paravermis holds the map of where the limb is now. Lost proprioception = can only move accurately while watching the arm, which is why B12 / posterior column loss mimics cerebellar disease. **Lateral zone = hands + planning, and the same machinery does cognition:** ordering errands on the way home (store first, it closes early, then gas, then liquor store). Eyes closed or alcohol stresses the flocculonodular / vestibular system, so you sway.

Region	Input	Damage
Vermis	Spinal proprioception	Impaired postural adjustments

Paravermis	Red nucleus, spinal proprioception; sensory maps that predict future limb position	Poor limb coordination
Lateral	Motor + association cortex via pons	Ballistic overshoot; poor motor learning and timing
Flocculonodular	Vestibular (canals, nuclei), visual (sup. colliculi, visual cortex)	Eye tracking, balance, gait disturbed

5. Syndromes

Syndrome	Signs	Classic cause
Posterior lobe (hemisphere) = arms and speech	Intention tremor, dysmetria, dysdiadochokinesia, scanning speech	Stroke, tumor, MS plaque
Anterior lobe	Lower-limb ataxia, wide gait, arms spared	**Alcoholic Purkinje cell degeneration**
Flocculonodular (midline)	Truncal ataxia, titubation, wide base, reels side to side, nystagmus	**Midline tumors in infants**

Why intention tremor is worst near the target: far away, small errors don't matter. Closing in, the cerebellum keeps yelling "too high, too low, too far left" and the cortex is slow to issue each fix, so the arm bounces between overcorrections. None at rest, mild at the start, worst at the end. Smart examiner holds the target far away and moves it.

Alcohol, 3 stories: (1) **Acute intoxication**: GABA receptors turned up, mossy/climbing systems disrupted, ataxic while drunk, normal when sober. (2) **Withdrawal**: opposite, excess excitation, tremor and jitters. (3) **Chronic: Purkinje cells are unusually vulnerable to alcohol** and die (granule cells too, less so); permanent; anterior lobe / legs first. His patient: sober in hospital for months, still cannot tandem walk. A drunk and a cerebellar patient look the same on the roadside heel-to-toe test.

Titubation child: toddler who sat fine starts reeling when sitting, over weeks to months. Sit, arms out, trunk sways = midline (fastigial / vermis / flocculonodular). Child + midline = tumor until proven otherwise; scan now. **Gait video:** teenager looks drunk, foot swings out and forward too far (no brake), cannot tandem walk, worse on the left, so lesion on the left.

Vocabulary and bedside test

Sign	Meaning	Test
Ataxia	Lack of coordination	Gait, heel-to-shin
Gait ataxia	Wide-based, staggering, "drunk"	Tandem walk
Truncal ataxia	Cannot hold trunk steady sitting	Sit unsupported, arms out
Titubation	Rhythmic bobbing of head/trunk	Observe at rest
Intention tremor	Worsens near target, absent at rest	Finger-to-nose
Dysmetria	Wavering, overcorrect, overshoot (past-pointing)	Finger-to-nose, heel-to-shin
Dysdiadochokinesia	Clumsy rapid alternating movements	Palm up / palm down on thigh
Scanning speech	Slow, syllables separated	Listen
Dysarthria	Slurred articulation	Listen
Nystagmus	Jerky rhythmic eye movements	Follow finger laterally

Hook (arm signs): three D's and a tremor: Dysmetria, Dysdiadochokinesia, Dysarthria, intention tremor.

6. Cerebellum vs basal ganglia

Cerebellar	Basal ganglia
Nystagmus, ataxia, dysarthria	**Hypokinesia**: Parkinsonism (akinesia, rest tremor, rigidity, lost postural reflexes)
Intention tremor	**Hyperkinesia**: chorea (rapid irregular), dystonia (twisting sustained posture), tics (shock-like face/neck/voice), tremor (to-and-fro around a joint, rest tremor in PD)
Titubation, dysdiadochokinesia, dysmetria	

The tremor question: ask "when is it worst?" Reaching = cerebellum. Resting, settles on reaching = basal ganglia. PD: tremor at rest, gone when they move. Cerebellar: the reverse, because an idle limb has nothing to coordinate.

7. Chiari malformation

Tonsils herniate through the foramen magnum into the upper cervical canal. Sagittal MRI: cerebellum sits below the skull base, crowding medulla and cord (normal drawing: cerebellum tucked in the posterior fossa). Gives headache (worse with cough), lower cranial nerve signs, cerebellar signs, and a **syrinx**. No spare room at the foramen: every pressure spike (child crying, coughing, straining) pushes the tonsils down a bit more, year after year, until they crush the upper cord and block CSF flow; the trapped CSF forms a syrinx in the cord (cape-like pain/temperature loss from the spinal cord lecture). "Easy to recognize on sagittal MRI."

8. Localize in 3 questions

1. **Cerebellum at all?** Ataxia, dysmetria, intention tremor, nystagmus with **normal strength and sensation**. Weakness = corticospinal. Ataxia much worse eyes closed + lost position sense (Romberg) = posterior columns / nerve. Looks cerebellar but could be the inputs (pons, MCP, B12).
2. **Midline or hemisphere?** Trunk and gait, limbs fine lying down = midline (vermis / flocculonodular / fastigial). Limb dysmetria + intention tremor = hemisphere (dentate side).
3. **Which side?** Same side as the limb signs.

9. Vignette decoder

Clue	Answer
Sudden right hand past-pointing + right dysdiadochokinesia, strength/sensation normal	Right cerebellar hemisphere (posterior lobe) infarct (PICA/SCA)
Toddler, weeks of morning vomiting/headache, reels when sitting, wide base, nystagmus, arms OK lying down	Midline tumor (medulloblastoma), flocculonodular/vermis; 4th ventricle block → hydrocephalus
Heavy drinker, sober 2 months, can't tandem walk, heel-to-shin bad, finger-to-nose fine	Anterior lobe syndrome, chronic Purkinje loss (not intoxication, not withdrawal)
MS, new left arm intention tremor + dysmetria, nystagmus on left gaze	Left cerebellar hemisphere plaque (or left SCP)
Shuffling, pill-rolling tremor that stops when reaching, rigid, masked face	NOT cerebellar: Parkinsonism (basal ganglia)
Hand wavers near the glass, overshoots, corrects, overshoots; still at rest	Intention tremor + dysmetria, ipsilateral hemisphere
Coronal section: big crinkled ribbon far lateral in white matter	Dentate (midline clump = fastigial; small pair between = globose + emboliform; bundle leaving upward = SCP)
Cough headache, cape pain/temp loss, low cerebellum on sagittal MRI	Chiari malformation with syrinx
Vegan, months of unsteady gait, lost vibration/position in feet, worse eyes closed, brisk reflexes, normal cerebellum MRI	B12 subacute combined degeneration: lost cerebellar INPUT, not cerebellar
Coarse arm shake lifting a water bottle, still in the lap	Rubral (red nucleus) tremor
Trunk sways sitting with arms out, limbs coordinated	Fastigial / midline (vermis, flocculonodular)
Teen "looks drunk" at 2 am, isn't; foot swings too far each step, worse on left	Gait ataxia, left cerebellar hemisphere

60-second self-quiz (cover the right side)

Main input peduncle?	Middle (pons)
Output peduncle, to where?	Superior → VL thalamus → motor cortex; also red nucleus
Only source of climbing fibers?	Inferior olive
Only output cell of cerebellar cortex?	Purkinje (inhibitory)
Only output of the cerebellum as a whole?	Deep nuclei
Nucleus for trunk posture?	Fastigial
Cells that die in alcoholics?	Purkinje
Herniates in Chiari, through what?	Tonsils, foramen magnum
Right cerebellar lesion, signs on which side?	Right (double cross)
Flocculonodular inputs?	Vestibular + visual
Cerebellar function that is not motor?	Lateral zone plans cognitive sequences
Strength and sensation in a pure cerebellar lesion?	Normal